

求证

$$\sum_{k=0}^n \frac{1}{(2k+1)!(2n-2k)!} = \frac{4^n}{(2n+1)!}$$
$$\Leftrightarrow \sum_{k=0}^n \binom{2n+1}{2k+1} = 4^n$$

同理有

$$\sum_{k=0}^n \binom{2n+1}{2n-2k} = 4^n$$

以及有恒等式

$$\sum_{k=0}^{2n+1} \binom{2n+1}{k} = 2^{2n+1}$$

现在可以得到

$$\sum_{k=0}^{2n+1} \binom{2n+1}{k} - \sum_{k=0}^n \binom{2n+1}{2n-2k} = \sum_{k=0}^n \binom{2n+1}{2k+1} = 4^n$$

从而原不等式得证